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CHM 101: General Chemistry I - Atoms, Bonding and Reactions

The structure-and-bonding foundations behind every chemistry course you will take

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Pure Chemistry | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

Study smarter. Ask better questions. Pass with understanding.

Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

CHM 101 rewards students who can move between the microscopic picture and the calculation. This pack connects atomic structure, periodicity, bonding and stoichiometry, then shows how to set up mole calculations that do not collapse under exam pressure.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe atomic structure, electronic configuration and periodic trends with correct reasoning.
- Predict bond type and molecular shape using electronegativity and VSEPR theory.
- Balance equations and solve mole, limiting-reagent and concentration problems.
- Explain the behaviour of acids, bases and buffers at an introductory level.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Atomic structure and periodicity

The periodic table is not a chart to memorise; it is a map of electron configuration.

- Atomic number is the number of protons; mass number is protons plus neutrons.
- Isotopes have the same atomic number but different mass numbers.
- Electrons fill orbitals in order of increasing energy, following the Aufbau principle, Pauli exclusion principle and Hund's rule.
- Atomic radius decreases across a period and increases down a group.
- Ionisation energy increases across a period and decreases down a group.
- Electron affinity and electronegativity follow similar trends to ionisation energy.

> **Explain, do not just state**
> A question asking why ionisation energy increases across a period wants the effective nuclear charge argument, not a restatement of the trend.

Bonding and molecular shape

Bonding is about electrons: transfer creates ions, sharing creates covalent bonds, and unequal sharing creates polarity.

- Ionic bonding: metal to non-metal, lattice structures, high melting points.
- Covalent bonding: shared electron pairs, discrete molecules or giant networks.
- Metallic bonding: delocalised electrons explain conductivity and malleability.
- Intermolecular forces: London dispersion, dipole-dipole and hydrogen bonding, in increasing strength.
- VSEPR: electron pairs repel and adopt the arrangement that maximises separation.
- Polar bonds do not guarantee a polar molecule; symmetry can cancel the dipoles.

Moles and stoichiometry

Almost every calculation in CHM 101 starts with converting something into moles.

1. Write and balance the equation first.
2. Convert given mass, volume or concentration into moles.
3. Use the balanced equation ratio to find the moles of the required substance.
4. Identify the limiting reagent when two reactant amounts are given.
5. Convert moles back to the requested unit: mass, volume of gas at stated conditions, or concentration.
6. Check significant figures and units.

> **The limiting reagent test**
> Divide the moles of each reactant by its coefficient. The smallest quotient identifies the limiting reagent - not the smallest mass.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. How many moles are in 18 g of water?

Answer: Molar mass of water is 18 g/mol, so 18 g is 1 mole.

2. Why is the first ionisation energy of sodium lower than that of magnesium?

Answer: Sodium's outer electron is in the 3s orbital and removing it gives a stable noble-gas configuration; magnesium's electron is held by a greater effective nuclear charge, so more energy is needed.

3. Predict the shape of methane and explain it.

Answer: Tetrahedral, because four bonding pairs around carbon repel equally and adopt the arrangement that maximises separation with 109.5 degree angles.

4. 20 g of calcium carbonate reacts with excess acid. Find the moles of carbon dioxide produced.

Answer: Molar mass of CaCO3 is about 100 g/mol, so 20 g is 0.2 mol. The reaction is 1:1, so 0.2 mol of CO2 is produced.

5. Is carbon dioxide polar?

Answer: No. The C=O bonds are polar but the linear geometry makes the dipoles cancel.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write electron configurations for the first 20 elements from memory.
- [ ] Sketch the periodic trends for radius, ionisation energy and electronegativity with reasons.
- [ ] Draw VSEPR shapes for five common molecules and label bond angles.
- [ ] Solve ten stoichiometry problems including three limiting-reagent cases.
- [ ] Make a one-page summary distinguishing ionic, covalent and metallic bonding.
- [ ] Perform a titration calculation from raw readings without a template.
- [ ] Attempt one past CHM 101 objective paper and review every wrong answer.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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